2048 game PORSCHE edition

Project submitted for the partial fulfillment of the requirements for the course

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Introduction

The 2048 Porsche Edition is an Artificial Intelligence based puzzle game inspired by the classic 2048 game, redesigned with a Porsche automotive theme to enhance visual appeal and user engagement. The project combines game development, graphical user interface design, and artificial intelligence techniques to create an intelligent game-playing agent capable of autonomously solving the puzzle. The primary objective of the game is to merge tiles of the same value until the target tile 2048 is reached. In this themed version, each tile value corresponds to a Porsche car model, making the gameplay more interactive and unique.

The project is implemented using Python programming language, where the graphical user interface is developed using the Tkinter library. The AI component of the project uses a Greedy Heuristic Search algorithm to evaluate possible moves and determine the best action at every game state. The heuristic function estimates the quality of the board based on tile positions, empty cells, and weighted scoring strategies.

This project demonstrates practical implementation of AI concepts in a real-world application and provides insight into heuristic search, state-space exploration, and intelligent decision-making in stochastic environments. It also showcases software engineering practices such as modular architecture, separation of concerns, and optimization for real-time performance.

Problem Statement

The purpose of this project is to develop an intelligent automated player for the 2048 puzzle game capable of making optimal or near-optimal decisions in a stochastic environment. Unlike deterministic games, 2048 introduces randomness after every move by spawning a new tile in an empty position. This makes the decision-making process significantly more complex, as the AI must choose actions without knowing the exact future state of the board.

Traditional brute-force search algorithms such as Breadth First Search and Depth First Search are impractical due to the exponentially growing state space and random tile generation. Therefore, a heuristic-based approach is necessary to estimate the quality of each board state and choose moves efficiently.

The main problem addressed by this project includes:

- Designing an AI that can make intelligent decisions under uncertainty.
- Maintaining large tiles in advantageous board positions.
- Preserving enough empty spaces to avoid dead-end states.
- Optimizing performance to allow real-time gameplay.
- Integrating AI logic with an interactive and visually appealing GUI.

The project aims to solve these challenges while producing a system that is efficient, scalable, and suitable for academic demonstration.

Methodology

The methodology adopted for this project follows a modular and iterative development approach. The system is divided into independent modules responsible for user interface, game logic, and AI decision making. Each module is designed and tested separately before integration.

Initially, the graphical layout of the game board is created using Tkinter. A 4×4 matrix is used internally to represent the board state. Game logic functions are then implemented to support tile movement in four directions, merging of adjacent equal tiles, and generation of random tiles after every valid move.

The AI module is developed using Greedy Heuristic Search. For every turn, the AI simulates all possible valid moves and computes a heuristic score for the resulting board state. The heuristic function uses weighted positional scoring and empty-cell counting to evaluate the board's quality. The move yielding the highest heuristic score is selected as the best move.

To improve efficiency, deep copies of the board are used during simulation so that the actual game state remains unaffected while evaluating hypothetical moves. Optimization strategies such as limiting simulation depth and sampling candidate positions are employed to maintain smooth runtime performance.

After implementation, the modules are integrated and tested repeatedly under multiple gameplay scenarios to verify correctness, performance, and AI effectiveness.

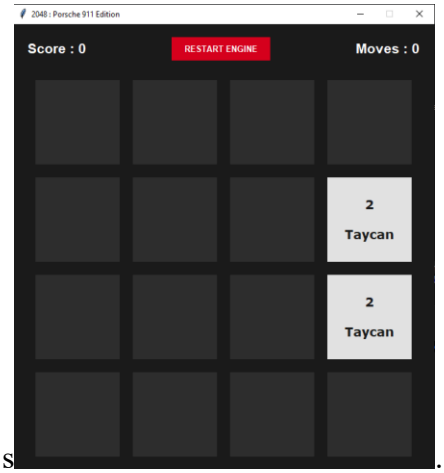
Architecture

The architecture of the 2048 Porsche AI Game describes how the system processes user/game input and produces intelligent moves using an AI-based heuristic search technique. It consists of three major layers: **Input Layer, Processing Layer, and Output Layer**, connected through a structured workflow.

Input Layer

The system accepts the current **4×4 game board matrix** as input, where each cell contains either a tile value (2, 4, 8, etc.) or 0 representing an empty space. The input may be generated either through player keyboard actions or through AI simulation during autonomous gameplay. The board state is continuously updated after every valid move.

The key responsibilities of the Input Layer are to read and store the board in a suitable data structure (2D array/matrix), identify all empty cells available for new tile generation, track

s 
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Processing Layer

This is the core of the system where the game logic and Artificial Intelligence are applied.

Game Logic Engine : Handles all core 2048 operations including tile movement, tile merging, score calculation, and generation of new random tiles after every valid move. It ensures the board follows all official 2048 rules.

Move Simulation Module

Creates temporary copies of the board using deep copy and simulates all possible moves (Up, Down, Left, Right) without affecting the real game state. This allows safe evaluation of hypothetical future board states.

Heuristic Evaluation Module

Calculates a heuristic score for each simulated board based on strategic tile placement and board quality.

It evaluates:

- Weighted tile positions using a weight matrix
- Number of empty cells available
- Overall board openness and future mobility

Greedy Search Controller

Compares heuristic scores of all valid moves and selects the move with the highest score as the optimal next action.

Game State Validator

Checks for terminal conditions such as:

- No valid moves remaining
- Board completely full
- 2048 tile achieved

Search Control Flow

Repeats the process recursively/iteratively until:

- The game reaches a terminal state
- The player stops the AI
- No further valid moves exist

- **Output Layer**

The output layer displays the updated board state after every move through the graphical user interface. It visually renders Porsche-themed tiles, updates the score and move counters, and shows the final game result when the game ends.

The key responsibilities of the Output Layer are:

- Display updated tile positions and values
- Show Porsche car model labels for each tile
- Update score and move statistics in real time
- Display win/lose conditions
- Refresh the GUI after every move

Example Workflow

1. Current board state is read from the game matrix.
2. AI simulates all four possible moves.
3. Heuristic score is calculated for each resulting board.
4. Best move is selected by Greedy Search Controller.

5. Game Logic Engine applies the move.
6. New random tile is generated.
7. Updated board is displayed to the user.
8. Process repeats until game over.

Algorithm (Greedy Heuristics Search)

The AI in this project uses the Greedy Heuristic Search algorithm to determine the next move. This algorithm is chosen because the 2048 game has a large state space and random tile generation, making exhaustive search computationally expensive.

At each turn, the AI performs the following steps:

1. Generate all possible legal moves (Up, Down, Left, Right).
2. Simulate each move on a temporary copy of the current board.
3. Apply the heuristic evaluation function to each resulting board.
4. Compare heuristic scores of all valid moves.
5. Select the move with the maximum score.
6. Execute the selected move in the real game.

The heuristic evaluation function is designed to maximize board quality by considering:

- Weighted positional scoring for tile placement.
- Bonus points for empty cells.
- Encouragement of monotonic tile ordering.

This approach allows the AI to make intelligent local decisions rapidly, even though it does not perform deep future-state analysis. While greedy search may not always produce globally optimal decisions, it provides a strong balance between performance and effectiveness for this project.

Test Cases

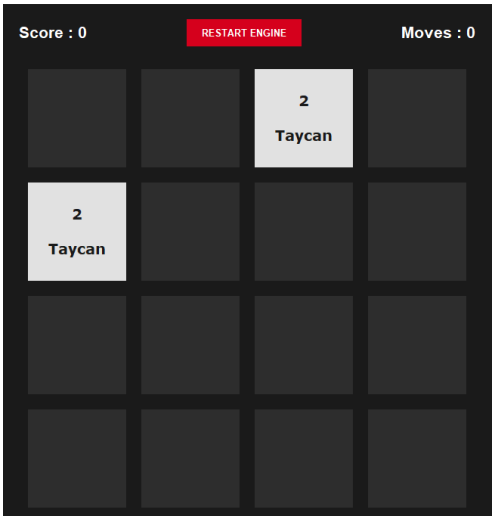
The system is tested using multiple scenarios to verify correctness and reliability of both gameplay and AI functionality.

Test Case 1: Game Initialization

Input: Start the application

Expected Output: Two random tiles should appear on the board

Result: Pass

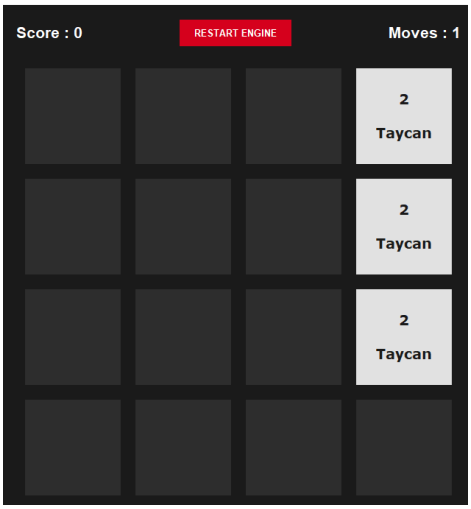


Test Case 2: Tile Movement

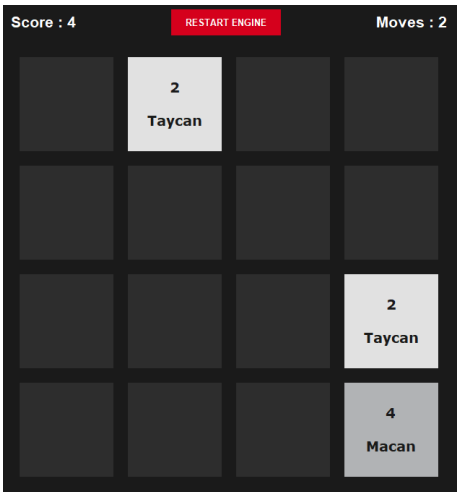
Input: Press Left Arrow

Expected Output: All tiles shift left according to game rules

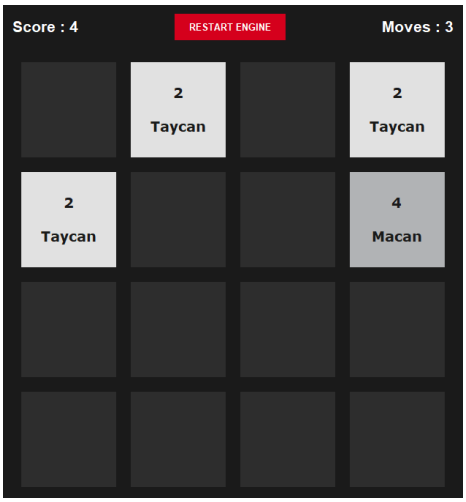
Result: Pass



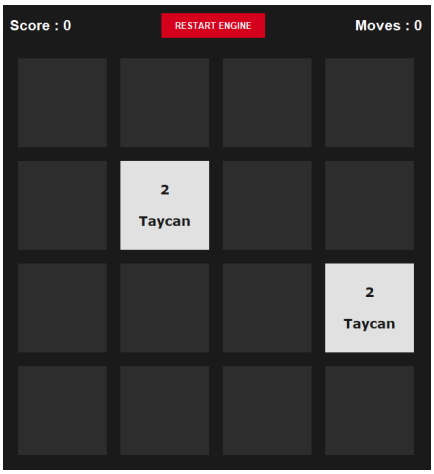
Test Case 3: Tile Merge
Input: Two adjacent equal tiles moved together
Expected Output: Tiles merge into one with doubled value
Result: Pass



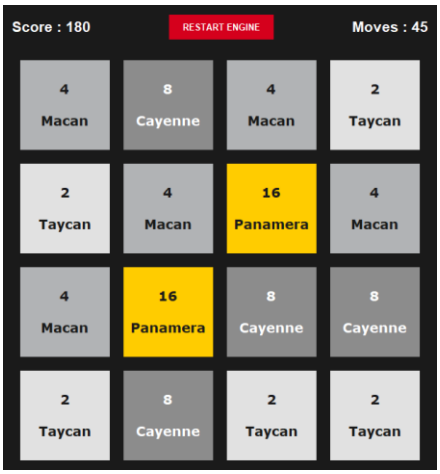
Test Case 4: AI Move Selection
Input: Trigger AI Move
Expected Output: AI chooses move with highest heuristic score
Result: Pass

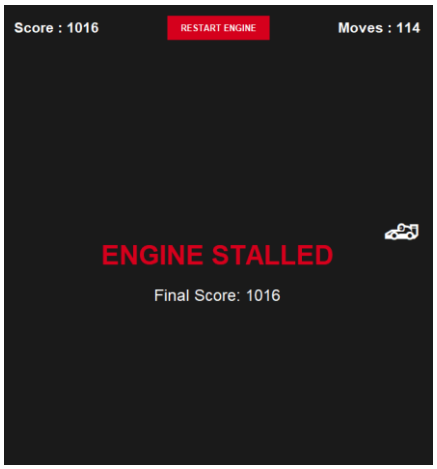


Test Case 5: Restart Button
Input: Click Restart
Expected Output: Game resets with fresh board
Result: Pass



Test Case 6: Full Board Detection
Input: Fill board with no valid moves
Expected Output: Game Over triggered
Result: Pass





All test cases confirm the system behaves as expected under normal and edge-case conditions.

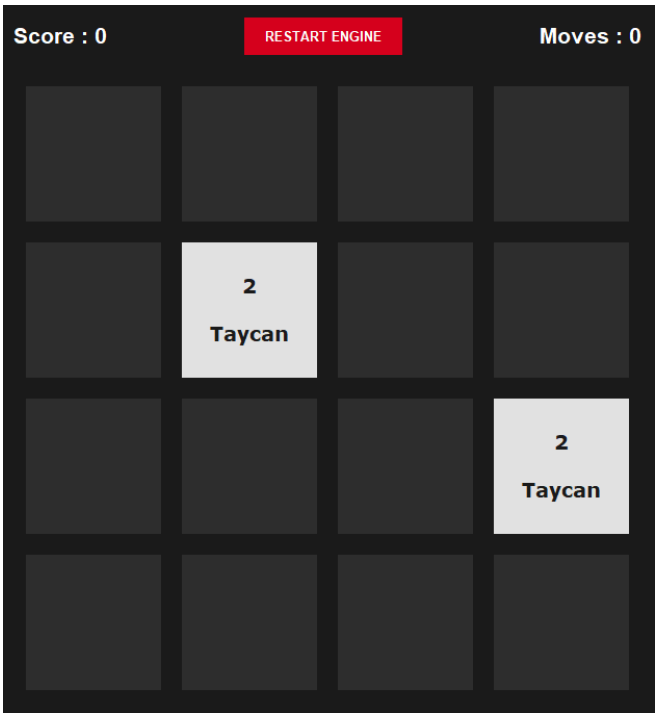
Output

The developed system successfully generates a playable Porsche-themed 2048 game with integrated AI automation. The user interface displays a responsive 4×4 game board with color-coded Porsche model tiles corresponding to numerical values. Real-time score and move counters update dynamically during gameplay.

The AI demonstrates intelligent move selection by consistently prioritizing strategic tile placements, maintaining board openness, and maximizing merge opportunities. During testing, the AI frequently achieved tile values of 512, 1024, and in several runs reached the 2048 tile.

The Restart Engine functionality allows instant resetting of the game state without restarting the application. The overall system provides smooth performance and stable execution across repeated gameplay sessions.

The project output confirms successful integration of GUI, AI, and game logic into a unified working application.



Conclusion

The 2048 Porsche Edition AI project successfully demonstrates the implementation of Artificial Intelligence techniques in an interactive puzzle game environment. By utilizing Greedy Heuristic Search, the AI is capable of evaluating multiple board states and selecting moves that maximize board quality in real time.

The project effectively addresses the challenges posed by stochastic tile generation and large search space through heuristic approximation and optimized simulation. It combines theoretical AI principles with practical software engineering concepts such as modular architecture, event-driven programming, and performance optimization.

In addition to solving the technical challenges, the project provides a visually appealing and user-friendly themed interface that enhances the gameplay experience. The modular design of the system also allows for future expansion, such as integrating more advanced search algorithms like Expectimax, improving heuristics, or adding animations and persistent high-score tracking.

Overall, the project demonstrates the successful application of heuristic search in intelligent game playing and serves as a strong example of AI-driven software development.

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